

# Pablo Ibáñez-Porras

— **PREDOCTORAL RESEARCHER AT CISA-INIA-CSIC**

Applying physics and mathematics to agrotechnology through mathematical modelling and GIS algorithms.

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# About me

Predoctoral researcher at the Animal Health Research Centre (CISA-INIA-CSIC), specialising in spatial epidemiology and real-time surveillance systems for emerging diseases under a One Health approach. I combine my background in Physics and Mathematical Modelling with GIS, data science and machine learning to develop early-warning systems for avian influenza and other transboundary diseases, including DiFLUision, ProtectIA and SMARTe.

7

Years of experience

11

Publications

32

Scientific events

954,5

Accredited training hours



02

# Education and Experience

2025 - actualidad

## PhD in Computer Engineering (ongoing)

University of Salamanca

Programme jointly supervised by [Irene Iglesias](#) (Animal Health Research Centre, CISA-INIA-CSIC) and [Ángel Martín del Rey](#) (University Institute of Fundamental Physics and Mathematics, IUFFYM-USAL).

Development and application of an integrated epidemiological risk-assessment model using mathematical and computational modelling to analyse the factors associated with the introduction and spread of HPAI in wild birds and estimate transmission risk to livestock holdings.

Doctorado

2026 – 2027

## **Development of a model for the spread of HPAI among wild birds within the Iberian Peninsula and identification of epidemiological links between cases through phylogenetic analysis.**

Animal Health Research Centre (CISA-INIA-CSIC)

Position funded through a management agreement with the Spanish Ministry of Agriculture, Fisheries and Food (2025–2027)

Research appointment

2024 – 2026

## **Development of models, viewers and applications for rapid risk assessment of emerging pathogens**

Animal Health Research Centre (CISA-INIA-CSIC)

Position funded by the European Partnership on Animal Health and Welfare (2024–2027)

Research appointment

2023 – 2024

## **Cost-benefit model of early wild-boar vaccination following African swine fever (ASF) outbreaks**

Animal Health Research Centre (CISA-INIA-CSIC)

Position funded by the VACDIVA project (2023–2024)

Research appointment

2022 – 2023

## **Enhancement of the DiFLUtion warning system for avian-influenza introduction by wild birds in Spain and development of ProtectIA to assess introduction risk in poultry holdings**

Animal Health Research Centre (CISA-INIA-CSIC)

Position funded through a management agreement with the Spanish Ministry of Agriculture, Fisheries and Food (2022–2023)

Research appointment

2021

## Master's Degree in Mathematical Modelling

University of Salamanca

MSc thesis: 'Epidemic-spread model between communities defined on complex networks considering population flow'. · [View project](#)

Academic education

2021

## Monte Carlo modelling of proton-beam interactions with different targets

Centre for Pulsed, Ultrashort and Ultraintense Lasers (CLPU)

Undergraduate internship

2020

## BSc in Physics

University of Salamanca

BSc thesis: 'Complex networks: description, analysis and applications'. · [View project](#)

Academic education

## Continuing education by specialism

**+388 h**

### GIS and remote sensing

ArcGIS Pro and Online, QGIS, Google Earth Engine, remote sensing and geospatial application development.

**Includes 1 attended activity**

**+179 h**

### AI, data and programming

Machine learning and deep learning for research, Python, Docker, statistical analysis and reproducible workflows.

**Includes 1 attended activity**

**+40 h**

### Mathematics and modelling

Numerical methods, differential equations and applied scientific simulation.

**+67,5 h**

### One Health and agrotechnology

Health-risk analysis, transboundary diseases, veterinary pathology and digitalisation of the agricultural sector.

**Includes 5 attended activities**

**+280 h**

### Research and professional skills

Scientific communication, literature review, qualitative research, languages and professional skills.

## 03 Publications

PUBLICATION • 2025

### **WiBISS: A tool to estimate avoided lost revenue of African swine fever wild boar vaccination at municipality level**

Spatial simulation of wild-boar vaccination strategies against African swine fever. The scenarios are translated into trade restrictions and potentially avoidable losses for pig producers.

Frontiers in Veterinary Science •  
doi:10.3389/fvets.2025.1667173 ↗

PUBLICATION • 2024

### **DiFLUision: A novel space-time alert system for HPAI.**

A spatio-temporal warning system integrating outbreaks, environmental conditions and wild-bird movements to provide weekly forecasts of HPAI introduction risk.

German Journal of Veterinary Research •  
doi:10.51585/gjvr.2024.4.0107 ↗

### Other publications and co-authored contributions

2026

#### **Development of a dashboard for monitoring and risk assessment of highly pathogenic avian influenza in Kazakhstan**

Central Asian Journal of Veterinary Science ↗

2026

#### **Molecular surveillance of flaviviruses in wild cetaceans from Brazil.**

Journal of Wildlife Diseases ↗

2025

#### **Risk Assessment of Highly Pathogenic Avian Influenza (HPAI) in Wild Bird Species in Kazakhstan.**

German Journal of Veterinary Research ↗

2024

#### **Novel epidemiologic features of highly pathogenic avian influenza virus A H5N1 2.3.3.4b panzootic: a review.**

Transboundary and Emerging Diseases ↗

2024

#### **Novel alpha-, beta-, and gammaherpesviruses in Neotropical carnivores of Brazil.**

Transboundary and Emerging Diseases ↗

2024

#### **Bacterial septicemia and herpesvirus infection in Antarctic fur seals (Arctocephalus gazella) stranded in the São Paulo coast, Brazil.**

Veterinary Research Communications ↗

2023

**Molecular Detection and Characterization of *Mycoplasma* spp. in Marine Mammals from Brazil.**

[Emerging Infectious Diseases ↗](#)

2023

**DashFLUboard: a web-based tool for assessing and predicting global spread of highly pathogenic avian influenza in near real time**

[EsriNews ↗](#)

2023

**Avian influenza: changing disease dynamics and how real-time surveillance systems can help us anticipate them**

[Avinews ↗](#)

## 04 Academic Activity

32

scientific events

20

as lead author or speaker

16

oral presentations

12

posters

10

co-authored contributions

Avian-influenza surveillance

Environmental antimicrobial resistance

African swine fever

One Health and interoperability

Spatial epidemiology and GIS

Wildlife and environmental health

## Highlighted international contributions as lead author

ORAL • EUROPEAN ONE HEALTH ASSOCIATION ANNUAL SCIENTIFIC MEETING 2026 (EOHA 2026) • MAY 2026

### **SMART-E: A Spatial Digital Framework for Standardized Environmental AMR Surveillance**

A spatial framework for standardising environmental antimicrobial-resistance surveillance. It integrates anthropogenic sources, territorial context and sampling points to enable reproducible comparisons across regions.

POSTER • EUROPEAN ONE HEALTH ASSOCIATION ANNUAL SCIENTIFIC MEETING 2026 (EOHA 2026) • MAY 2026

### **From Science to Policy: Co-developing a Real-Time One Health Early Warning System for Avian Influenza (DiFLUtion)**

Evolution of DiFLUtion into an operational early-warning system co-developed with surveillance authorities. The contribution connects spatio-temporal modelling, user needs and translation into health policy.

POSTER • SVEPM CONFERENCE 2026+ • MARCH 2026

### ASF FrontWave: A Rapid Risk Assessment interactive mapping tool

A rapid-assessment viewer that interprets the speed and direction of the African swine fever epidemic front. It supports territorial prioritisation of surveillance and control measures.

POSTER • GARA SCIENTIFIC MEETING • APRIL 2025

### WiBISS: A tool to estimate economic benefits of African swine fever wild boar vaccination for pig producers

Spatial simulation of wild-boar vaccination strategies against African swine fever. The scenarios are translated into trade restrictions and potentially avoidable losses for pig producers.

POSTER • IAAAM 55TH ANNUAL CONFERENCE • MAY 2024

### MarineStrandingViewer: An Online Tool to Record Marine Mammal Strandings

A web tool for recording and visualising marine-mammal strandings. It centralises georeferenced observations and supports their exploration for research and surveillance.

## Other lead-author contributions

2025

### Mapping anthropogenic sources of antimicrobial-resistance emissions

Oral • Esri Spain Conference 2026

2025

### ProtectIA: A geospatial tool for avian-influenza prevention

Oral • 4th International Young Researchers Meeting INNOVA-Salamanca

2025

### Geoprocessing tools to support emerging-disease response

Oral • 1st Interdisciplinary Congress on Medicine and Engineering

2025

### DiFLUision: a tool for surveillance against Avian Influenza

Oral • 2nd General Assembly of the CSIC Computational Biology and Bioinformatics Network (Conexión BCB)

2024

### Epidemiology and spatial analysis: using GIS to understand disease dynamics and manage public and animal health from a One Health perspective

Oral • Esri Spain Conference 2024

2024

### Soil contamination by antibiotics and antimicrobial-resistance genes

Poster • 10th CONDEGRES Symposium

2023

### Real-time avian-influenza warning system: DiFLUision

Poster • Health Surveillance Meeting 2023: present and future towards 2030

2023

### Ants, chickens and mathematics

Oral • INIAciando Science

2022

### DiFLUision avian-influenza surveillance system: direct use of data from SEO/BirdLife citizen-science programmes

Poster • Spanish Ornithology Congress



## Scientific and technical advice

### Technical advice on the highly pathogenic avian influenza (HPAI) situation

• sessions: July 2022, February 2023, July 2023 and October 2023.

## 05 Taught courses and teaching activity

WORKSHOP LINGLOBAL SCIENTIFIC AND  
COMMUNICATION TRAINING • INSTRUCTOR • 2025

### The overlooked role of Passeriformes in avian influenza and antimicrobial resistance amid climate change

Workshop on the role of passerines in the  
connectivity of avian influenza and antimicrobial  
resistance under climate-change scenarios.

WORLD ORGANIZATION FOR ANIMAL HEALTH (WOAH)  
SUB-REGIONAL REPRESENTATION FOR SOUTH-EAST  
ASIA • INSTRUCTOR • 2025

### Training course on Risk analysis for transboundary animal disease control

International course on risk analysis for  
transboundary animal-disease control, delivered for  
WOAH's Sub-Regional Representation for South-East  
Asia.

WORKSHOP ON ARTIFICIAL INTELLIGENCE, BIG DATA AND  
DIGITALISATION IN VETERINARY MEDICINES •  
INSTRUCTOR • 2025

### Implementing AI in animal-health research

Applied training in artificial intelligence, big data and  
digitalisation for animal-health and veterinary-  
medicine research.

SCIENCE OUTREACH • INSTRUCTOR • 2024

### Mentor for the Scientists in Practice programme

Mentoring students during an immersive scientific-  
research placement at CISA-INIA/CSIC.

EXTERNAL-PLACEMENT SUPERVISION • INSTRUCTOR •  
2023

### Analysis of the spatio-temporal distribution of several animal diseases

Supervision of external placements focused on the  
spatio-temporal distribution of animal diseases.

BACHELOR'S THESIS SUPERVISION • INSTRUCTOR • 2023

### Mathematical tools for temporal smoothing of migration routes

Supervision of a bachelor's thesis on mathematical  
tools for temporal smoothing of migration routes.

TEACHING • 2022

### Analysis of the spatio-temporal distribution of HPAI outbreaks in Spain

Supervision of external placements on the spatio-  
temporal distribution of highly pathogenic avian-  
influenza outbreaks in Spain.



# Seminars, workshops and other forums

Lead

**Artificial Intelligence: foundations and models for the Epidemiology and Environmental Health group**  
CISA-INIA/CSIC (internal seminar) · June 2026

Lead

**ProtectIA: A multi-criteria tool for prioritising avian-influenza risk in livestock districts**  
Zoonotic Influenza Challenge Scientific Meeting · May 2026

Lead

**Data analysis in epidemiology**  
UII webinar series · November 2025

Lead

**DiFLUision: a GIS model applied to the livestock industry**  
Seminar · May 2024

Lead

**SMARTe: A spatial interface for standardised environmental AMR surveillance**  
CSIC Geographic Technology Meeting 2026: Innovation, Analysis and Science · May 2026

Instructor

**The overlooked role of Passeriformes in avian influenza and antimicrobial resistance amid climate change**  
Workshop LINCGLOBAL Scientific and Communication Training · December 2025

Instructor

**Implementing AI in animal-health research**  
Workshop on Artificial Intelligence, Big Data and digitalisation in veterinary medicines · February 2025

Lead

**DiFLUision: a real-time avian-influenza warning system**  
Central Veterinary Laboratory of Algete · April 2023

## 06 Developed tools

01

**DiFLUision**  
SPATIO-TEMPORAL WARNING SYSTEM FOR HIGHLY PATHOGENIC AVIAN INFLUENZA  
Identifies areas at risk of HPAI introduction in Spain each week. It integrates WOAHI notifications, AEMET temperatures, phenology and bird-migration routes. Its modular ArcGIS Pro and Python architecture generates operational surveillance alerts.  
[Open reference ↗](#)

02

**SMART-E**  
SURVEILLANCE AND MONITORING OF ANTIMICROBIAL RESISTANCE IN THE ENVIRONMENT  
A spatial digital framework for standardising environmental pressure associated with antimicrobial resistance. It integrates anthropogenic sources, territorial context and sampling points, enabling comparable and scalable AMR analyses across studies, regions and countries.  
[Open reference ↗](#)

03

## WiBISS

### WILD BOAR IMMUNIZATION STRATEGY SIMULATOR

Simulates wild-boar vaccination scenarios against African swine fever. It translates spatial effects into restriction zones and avoided losses for the pig sector, combining cellular automata, epidemiological data and economic assessment.

[Open reference ↗](#)

04

## ASF FrontWave

### AFRICAN SWINE FEVER FRONT-WAVE RAPID RISK ASSESSMENT

An interactive viewer for analysing the speed and direction of the ASF epidemic front in wild boar. It turns spatio-temporal notifications into a rapid picture of disease progression and supports the prioritisation of surveillance and territorial control strategies.

[Open reference ↗](#)

05

## Dashboards

### EPIDEMIOLOGICAL DASHBOARDS FOR DISEASE SURVEILLANCE AND RISK ASSESSMENT

Web dashboards integrating outbreaks, movements, susceptible populations and environmental variables to describe and anticipate risk. They include DashFLUboard and HPAI-surveillance applications for Europe and Kazakhstan, making spatial analysis accessible to non-specialist users.

[Open reference ↗](#)

## 07 Science outreach

### FESTIVAL PINT OF SCIENCE • 2025

#### Epidemic detectives: chickens and mathematics

Talk on how physics, mathematics and data can help track and anticipate an epidemic.

[Open link ↗](#)

### SCIENCE OUTREACH • 2024

#### Mentor for the Scientists in Practice programme

Mentoring students so they can experience scientific work and research methods first-hand.

### SCIENCE AND TECHNOLOGY WEEK • 2022

#### Epidemics and pandemics: become the most dangerous virus

An interactive activity exploring the spread and transmission chains of infectious agents through play.

[Open link ↗](#)

SCIENCE AND TECHNOLOGY WEEK  
• 2022

### Become a real epidemic detective

An interactive workshop focused on observation, data and decision-making during outbreaks.

EUROPEAN RESEARCHERS' NIGHT  
• 2022

### Epidemics and pandemics: become the most dangerous virus

An interactive activity exploring the spread and transmission chains of infectious agents through play.

[Open link ↗](#)

## 08 Awards, fellowships and research stays

**2026**

### MOVTEC 2025-12 research stay

Barcelona · February 2026

**2024**

### 2024 Research Award

Madrid · October 2024

**2021**

### Introduction to Research Fellowship

University Institute of Fundamental Physics and Mathematics (IUFFYM) · Salamanca · January 2021

## 09 Skills and Languages

**Python**

**MATLAB**

**Mathematica**

**Fortran**

**QGIS / ArcGIS**

**LaTeX**

**Office**

**English**

Language

## 10 Contact

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